Precision-fermented dairy, egg and serum proteins: what they cost, and where evidence would help

Memo · v4, seven proteins (2026-10-05 21:57) · Pablo Antonio Moreno Casares · techno-economic model `pf-tea`, Humbird (2021) method, checked against NREL, Intelligen and Voutilainen designs · prepared with an AI assistant; not yet reviewed by human experts

Bottom line.

- **Today's public strains are far too expensive.** At the best peer-reviewed titers, β-lactoglobulin (BLG, whey), ovalbumin (OVA), ovomucoid (both egg white) and β-casein cost $43.0, $227, $479 and $803 per kg protein. Whey protein, dried egg white and casein traded at $9–$18, $10–$21 and $8–$19 per kg protein in 2015–2025 (2025 dollars). The one public titer near the design point is a company's own: EVERY's 30 g/L ovomucoid patent example, at $23.7.
- **At the titers companies' cost models assume, the commodity proteins near their markets.** At 42 g/L BLG costs $15.8 ($12.3 with the lighter accounting of NREL-style design reports), OVA $15.9, ovomucoid $16.0 and β-casein $16.4; 3 of 21 annual whey export prices in 2015–2025 were above the BLG figure. Casein has the thinnest evidence: no company discloses a titer, and no secreted casein titer is public for any yeast or fungus.
- **The gate is secretion, not titer.** $20/kg needs about 5–7 mg protein per g of cells per hour at the base cell density (the base design: 4.7). With today's biology and capital, design alone reaches $7.1/kg BLG at 40 mg/g/h and $12.0 at 10. No published **K. phaffii** process is near either rate: the best glucose fed-batch we found reached 2.1, and in the one near-zero-growth study secretion fell as growth slowed.
- **Sugar is not the problem.** Base-case glucose ($4.1/kg BLG) costs no more than animal feed per kg of protein ($4.7–$6.7, US 2025, our derivation); the gap is the plant ($17/kg against $2–$8 of non-feed farm cost).
- **Best case, not a floor.** With every assumed limit at its best, 400 m³ vessels and 100 kt/yr, BLG, OVA, ovomucoid and β-casein cost $4.1, $4.2, $4.3 and $4.5 per kg if cells secrete 40 mg/g/h; at 10 mg/g/h BLG costs $8.1, and at the best rate demonstrated (2.1 mg/g/h) $24.5.
- **Scarce proteins are the easy case.** At its design titer (10 g/L) and a market-sized plant (250 t/yr), PF lactoferrin costs $130–$168/kg, below the $250–$1,585/kg reported for the dairy product; Intelligen's own model of the same route costs 2.9× ours to produce.
- **The route can change the product.** Precipitated, as dairy casein is, β-casein reaches dairy purity at $22.0/kg; the only public yeast casein process (membranes only) makes a powder that is about 6% β-casein, mostly yeast carbohydrate.
- **Albumin for cultivated-meat media: the columns set the cost.** With the industrial process's minimal train (cation exchange, then anion exchange), at cell-culture purity and 1 kt/yr, albumin costs $215/kg ($205 at 42 g/L; $424 at the best public titer, 8.7 g/L of broth made with methanol), against $40–$70 for bulk bovine albumin and GFI's $10/kg target. Capture alone would cost $143 and a $100/L resin $114; membranes alone ($34.3) reach only 77% albumin. At the base cost it adds $1.95–$10.23 per kg of cultivated meat, while leading firms are removing albumin from their media.
- **α-Lactalbumin and κ-casein: the strains are missing.** α-Lac (infant-formula whey protein) costs $33.8/kg at the base titer on a 1 kt/yr plant, against $25–$129 reported per kg α-lac (grade C), but $1,848 at its best public titer (0.6 g/L of supernatant, human α-lac). Rennet cheese needs κ-casein at about 19% of its casein: at the best κ titer (0.25 g/L) an α_{s1} + κ blend costs $567–$573/kg, and κ co-precipitated with α_{s1} must reach roughly α_{s1}'s design titer to add under 10%. At 0.45% use, heme from the best secreting strain adds $1.09 per kg of beef analogue at the factory gate.

$/kg protein, US	BLG	OVA	ovomucoid	β-casein	lactoferrin	albumin	α-lac
best public academic strain	$43.0	$227	$479	$803	$413	$424	$1,848
base case (30 g/L; lactoferrin 10 g/L)	$21.2	$21.3	$21.3	$22.0	$168	$215	$33.8
base case, design-report capital	$16.4	$16.5	$16.5	$17.0	$130	$185	$25.0
industry benchmark (42 g/L)	$15.8	$15.9	$16.0	$16.4	$106	$205	$28.2
physical-limit best case	$4.1	$4.2	$4.3	$4.5	$56.9	$172	$5.2
incumbent, 2025$ per kg protein	$8.7–$17.8	$9.7–$21.2	$9.7–$21.2	$7.9–$18.7	$250–$1,585	$40.0–$70.0	$25.0–$129

Plants: 10 kt/yr (lactoferrin 250 t/yr, albumin and α-lac 1 kt/yr); physical limit at 100 kt/yr (lactoferrin 1.6 kt/yr, α-lac about 19 kt/yr). Albumin's incumbent: bulk bovine albumin (technical grade).

Where evidence would help most, in our judgment:

1. **Secretion without growth.** How much protein a strain secretes per g of cells per hour when it barely grows, over days, at 1–10 m³. It sets both today's cost and the best case, for every protein here.
2. **Open fed-batch data.** Titer, productivity and sugar yield measured **together**; for casein, any secreted titer above a few g/L, and whether the unphosphorylated protein makes cheese like dairy casein; for bovine α-lac and κ-casein, any secreted titer above 1 g/L.
3. **Real plant costs.** Capital, staffing and running costs of recent PF plants would settle the vessel price basis and the fixed-cost rule (aseptic vessels would add $9.8/kg; the two accounting conventions differ by $4.7/kg on BLG).
4. **The design ceiling.** Workable starting volumes, feed concentrations and O_2 transfer at scale.

Animal welfare. The cost model does not say which protein matters most for animals, but the products differ: whey is a by-product of cheese, so displacing it mostly lowers the whey price; casein is what milk is bought for, and egg white (OVA, ovomucoid) carries far more animal-years per kg of protein than either.

EU. At Dutch sugar, energy, wage and construction prices the BLG plant costs about $3.1/kg more; a capital charge at 4% over 20 years instead of 7.5% over 10 would cut $3.9/kg. Cost effects on one plant, not policy recommendations.

Caveats. A production cost with a 14.6%/yr capital charge, not a market price. Excludes R&D, licensing, regulatory, sales, start-up, failed batches and first-of-a-kind premiums. Strain parameters rest on design values and company-reported averages (evidence grades B–C); casein and lactoferrin titers are borrowed from other proteins or species. Ovomucoid and β-casein were added in version 3, albumin, α-lactalbumin and the casein and heme notes in version 4; they have been audited by AI reviewers only (docs/AUDIT.md). Full report, audit log, simulated expert reviews and data: `pf-tea` repository.